

Ethanol Blending in Gasoline - Peru

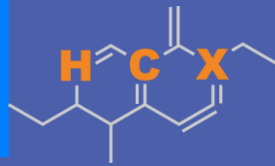
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Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.



Ethanol Blending in Gasoline - Peru



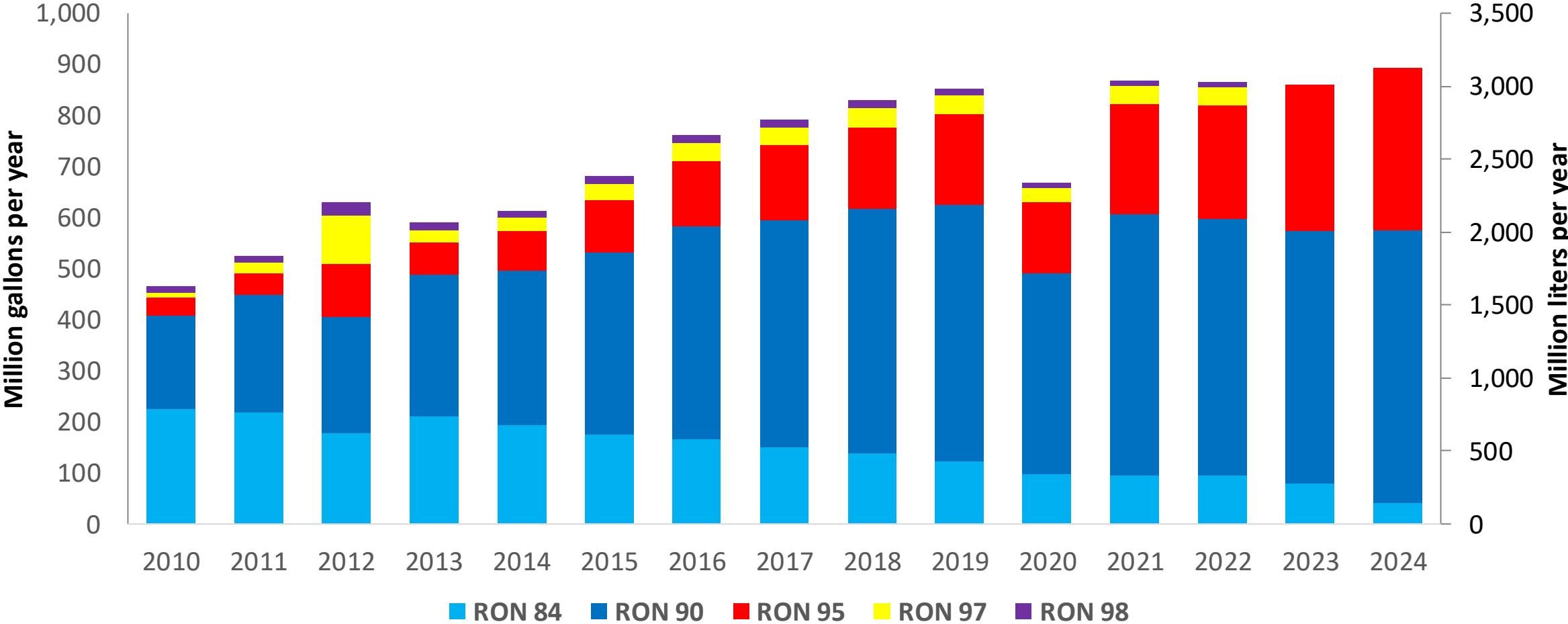
In 2024, Peru gasoline consumption surpassed 900 million gallons (3,000 millions liters). Starting in 2023 a program to increase octane in gasoline was implemented; RON84 and RON 90 gasoline are commercialized as Regular gasoline (RON 91) with a 70% market participation. RON 95, RON 97 and RON 98 gasolines are now marketed as Premium gasoline (RON 96) with a 30% market share. Gasoline with ethanol (gasohol) is required to meet one additional octane number (RON 91 and RON 96 respectively). Local gasoline supply is done with production from Petroperú, Repsol and Primax. Gasoline imports account for 10 to 15% of demand, and are supplied by local markets, United states and Europe.

Blends with 7.8% v/v are allowed (Decreto Supremo N.° 021-2007-EM: MINEM). Peru produces ethanol from sugarcane; however, its destination is mainly the EU market. Ethanol imports from the United States and to a lesser degree from Brazil are used for the national blend.

Source: MEM, 2024

Gasoline Demand in Peru

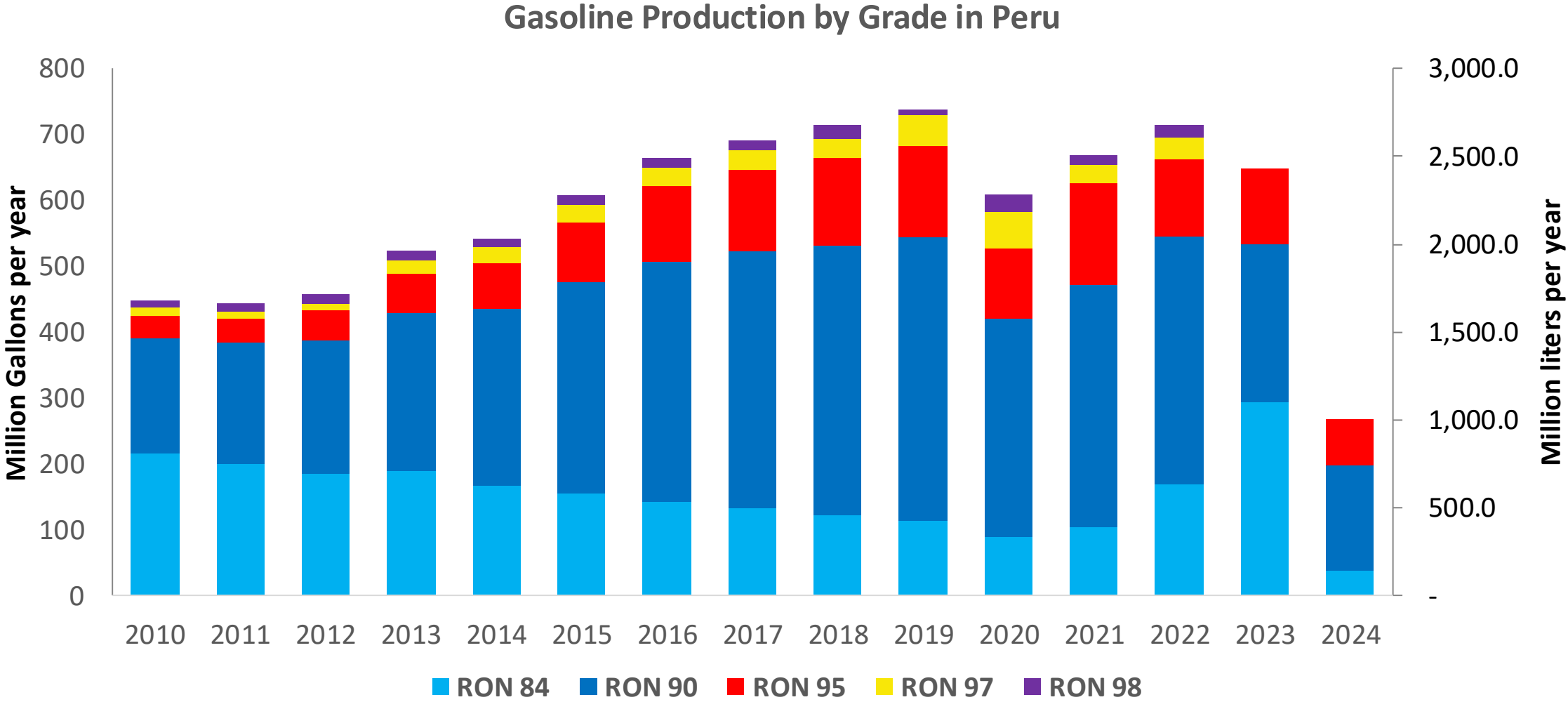
Gasoline Demand by Grade in Peru



Since 2023, RON 90 is commercialized as Regular Gasoline (RON 90-91) and RON 95, RON 97 and RON 98 are now marketed as Premium Gasoline (RON 95-96).

Source: MINEM, 2024

Gasoline Production in Peru

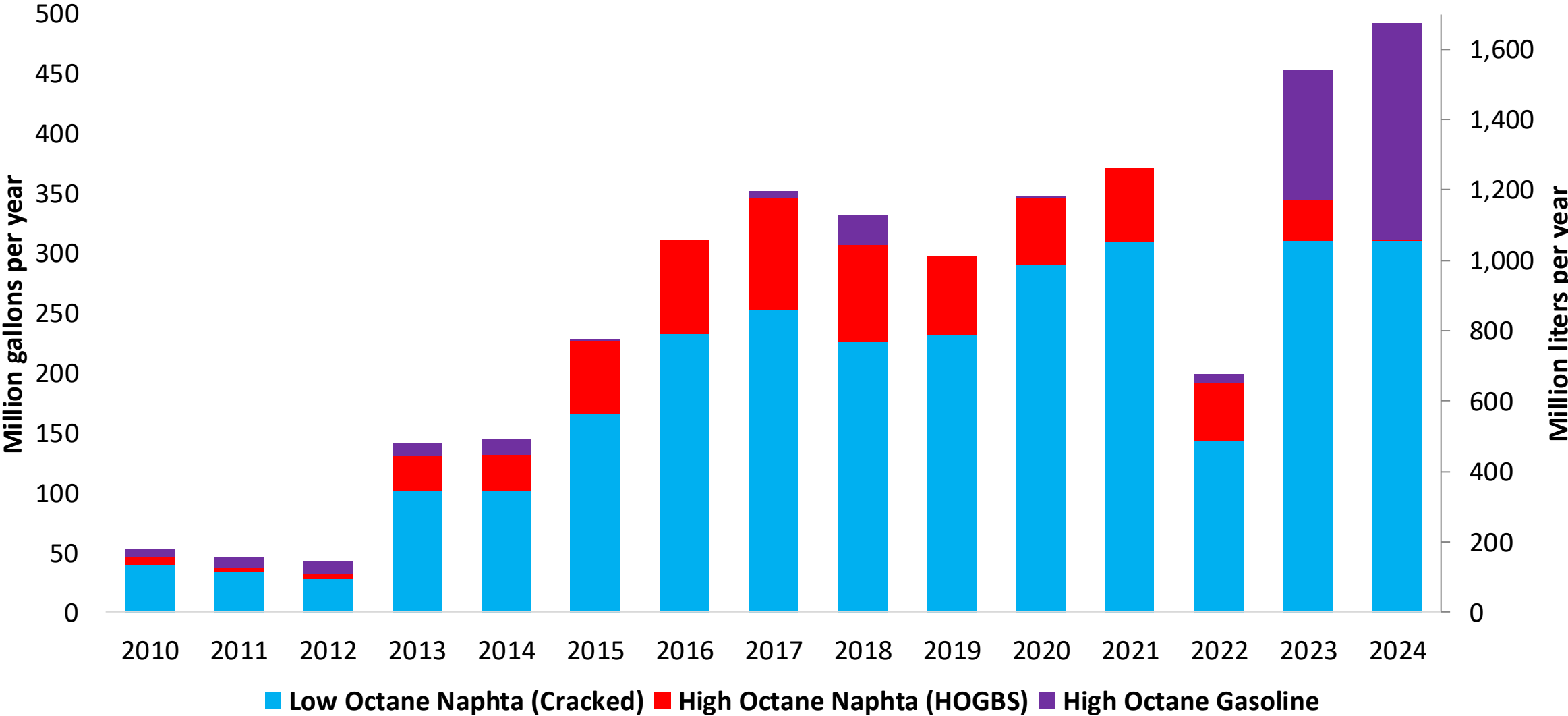


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Source: MINEM, 2024

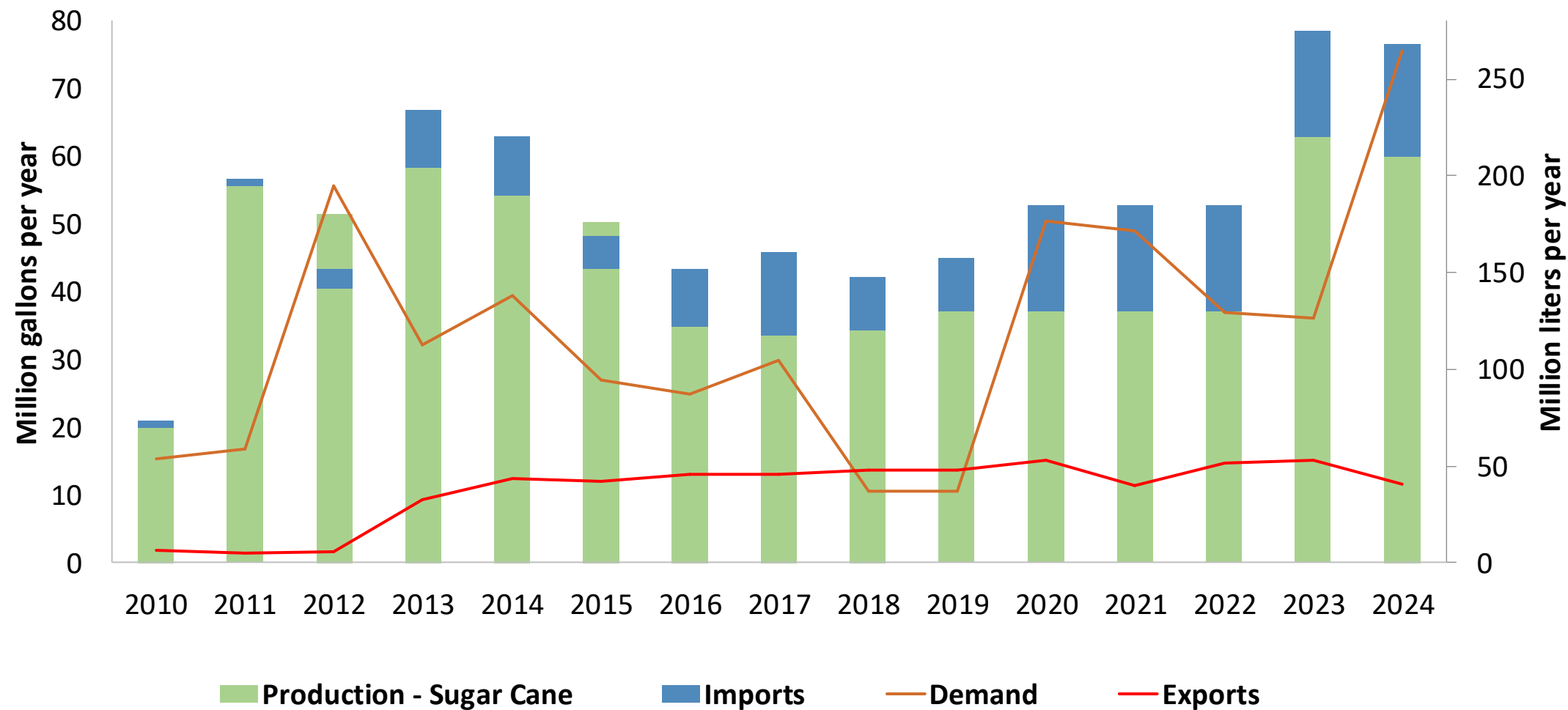
Gasoline Imports in Peru

Naptha and Gasoline Imports to Peru



Source: MINEM, 2024

Production and Demand of Ethanol in Peru



Source: MINEM, 2024

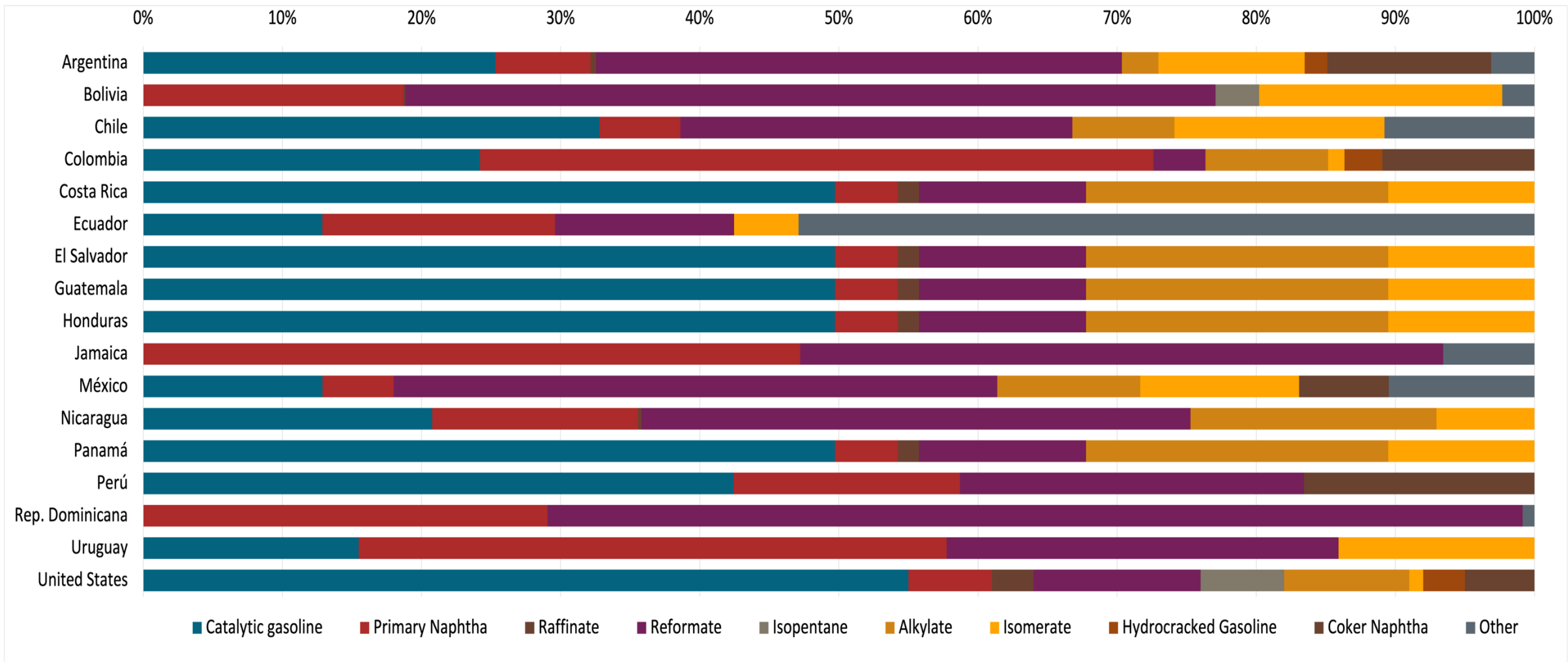
Gasoline Quality in Peru

Name	Ministerial Resolution N° 469-2021-MINEM/DM				EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2021				2017			
Applicability	Whole contry	Whole contry	Whole contry	Whole contry	All countries			
Selected Grade	Gasolina Regular	Gasolina Premium	Gasohol Regular	Gasohol Premium	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 2% v/v	< 2% v/v	< 2% v/v	< 2% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 45% v/v	< 45% v/v	< 45% v/v	< 45% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 25% v/v	< 25% v/v	< 25% v/v	< 25% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 0,25 mg/l	< 0,25 mg/l	< 0,25 mg/l	< 0,25 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 90	> 95	> 91	> 96	> 95	> 95	> 98	> 98
MON	-	-	-	-	> 85	> 88	> 85	> 88
AKI								
Sulfur Content	< 50 mg/kg	< 50 mg/kg	< 50 mg/kg	< 50 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	Reportar	Reportar	< 3,45 %m/m	< 3,45 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)			< 7,8 %v/v	< 7,8 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 76 kPa	< 76 kPa	< 76 kPa	< 76 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)								
RVP 37.8°C (Transition)								
MTBE					-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Source: MINEM, 2023

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



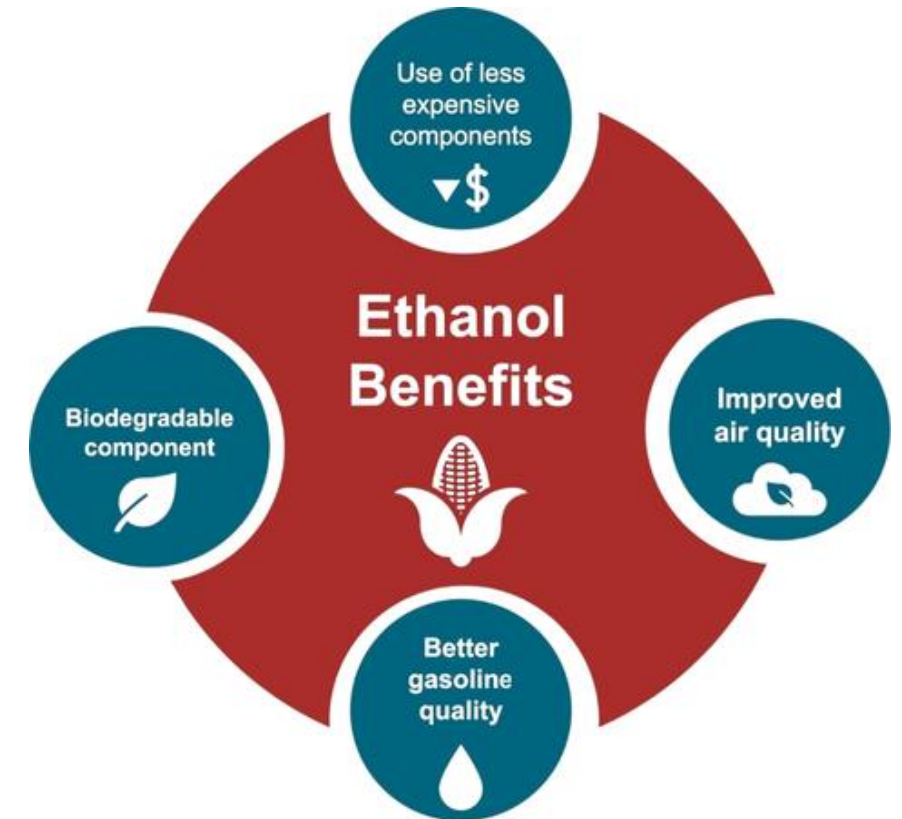
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

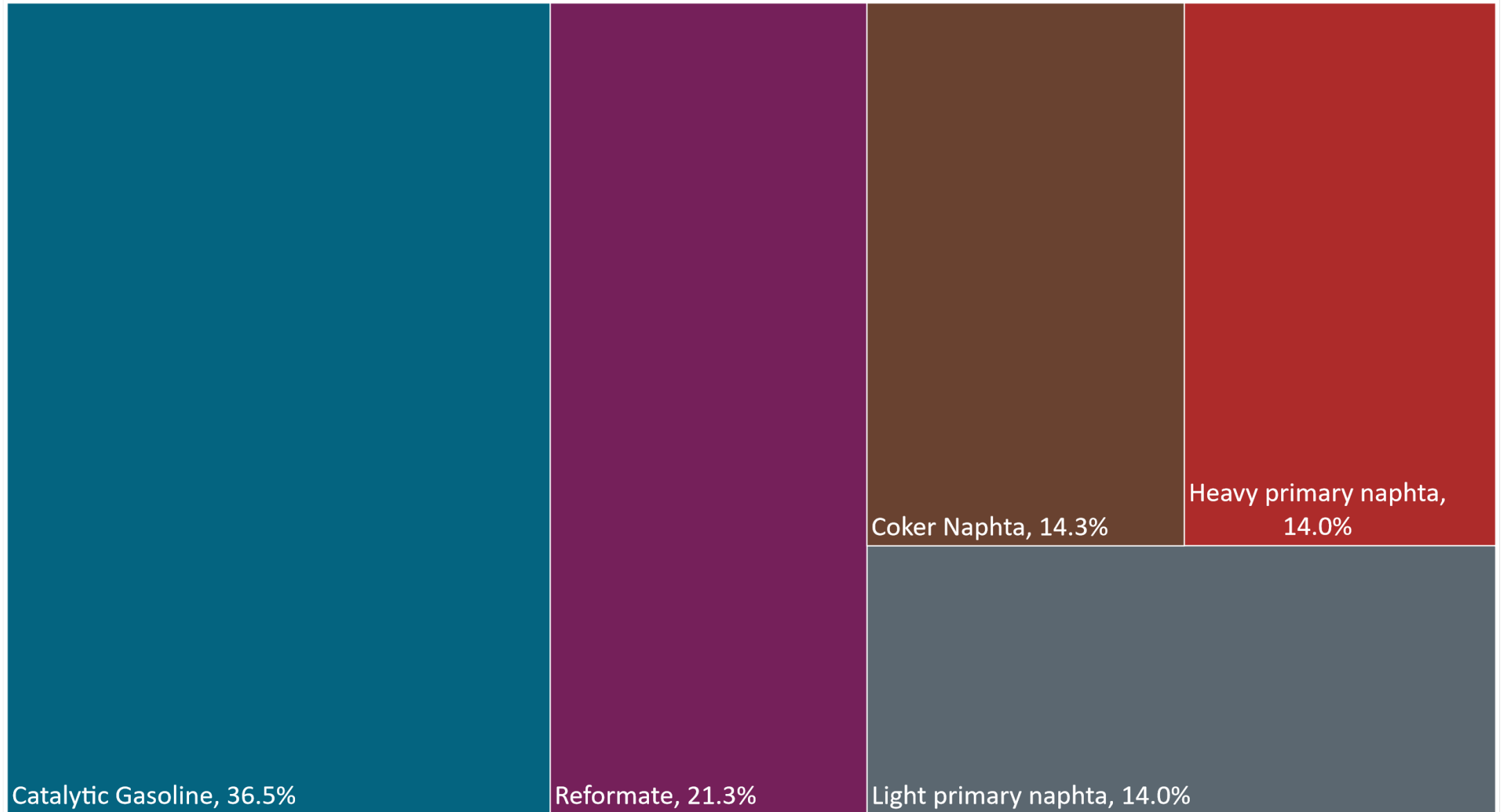
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

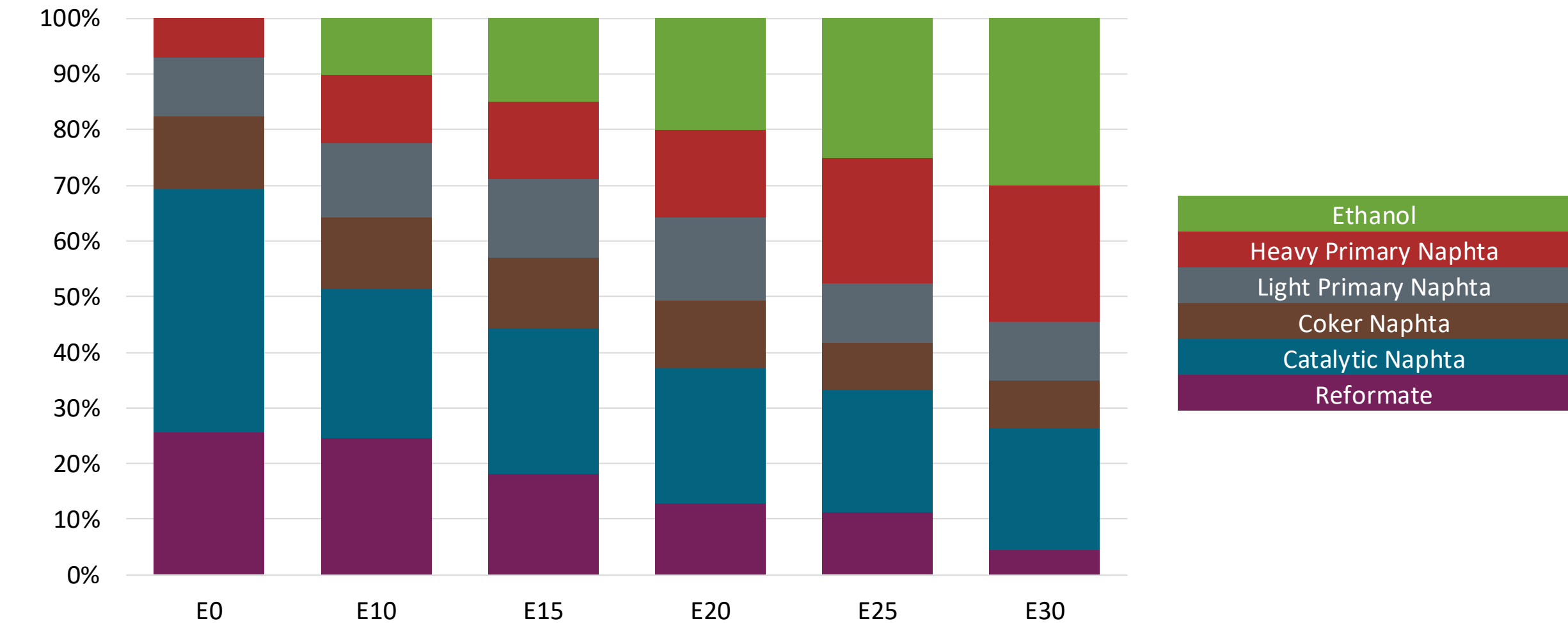
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Available Blending Components



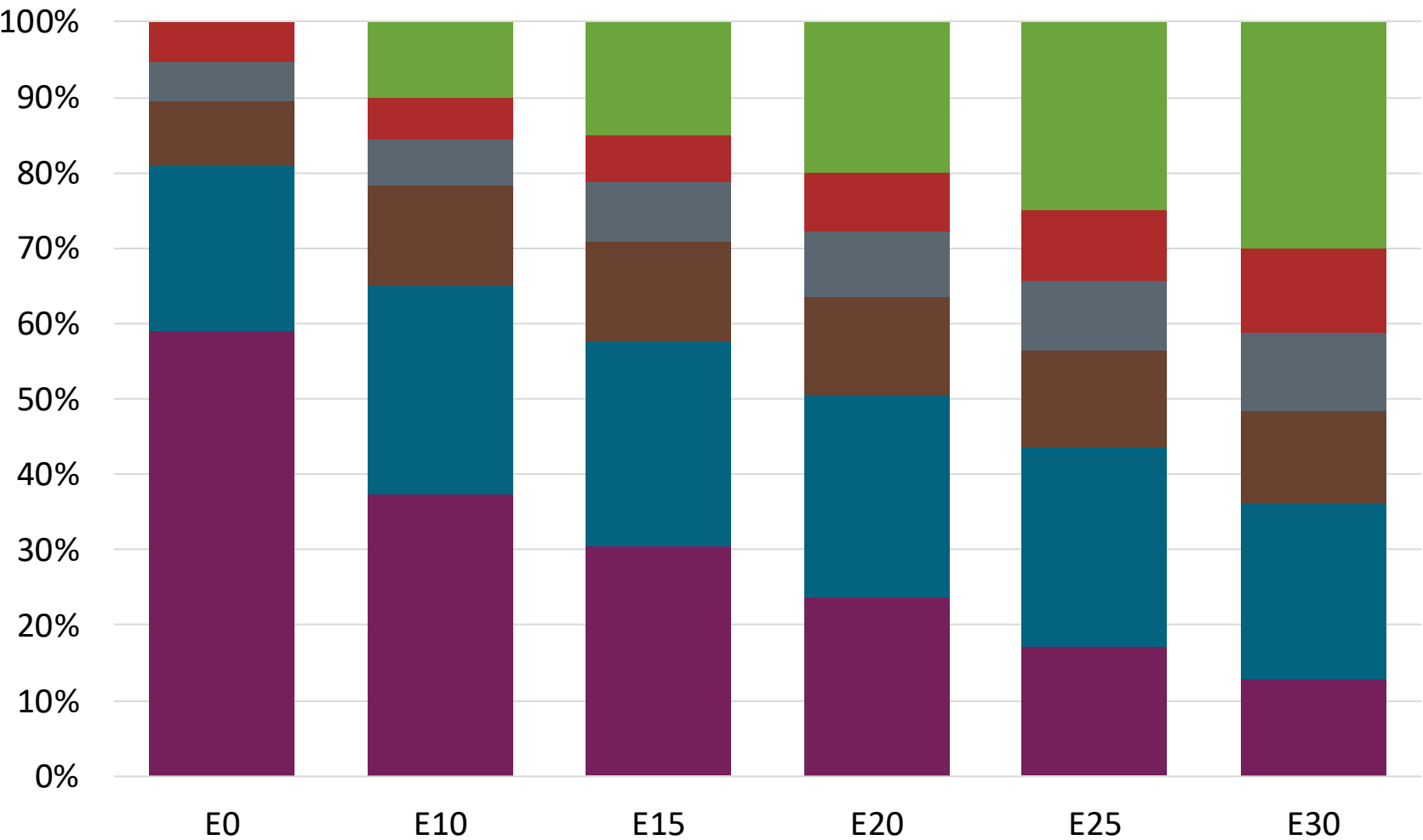
Peru – Regular – Constant Octane



Octane (RON)	90.0	90.0	90.0	90.0	90.0	90.0
Price (US\$/gal)	\$ 2.412	\$ 2.242	\$ 2.155	\$ 2.070	\$ 1.995	\$ 1.908

Source: Faro90

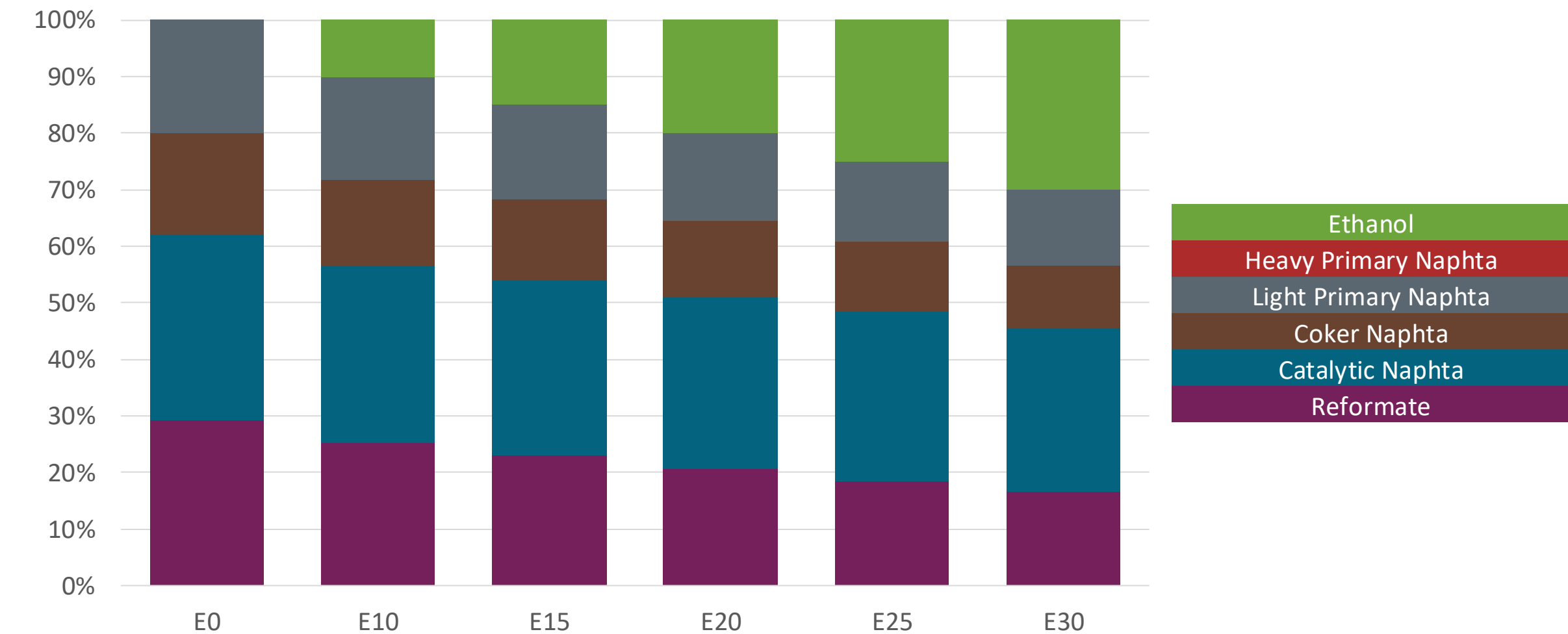
Peru – Premium – Constant Octane



Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (US\$/gal)	\$ 2.672	\$ 2.483	\$ 2.395	\$ 2.309	\$ 2.223	\$ 2.138

Source: Faro90

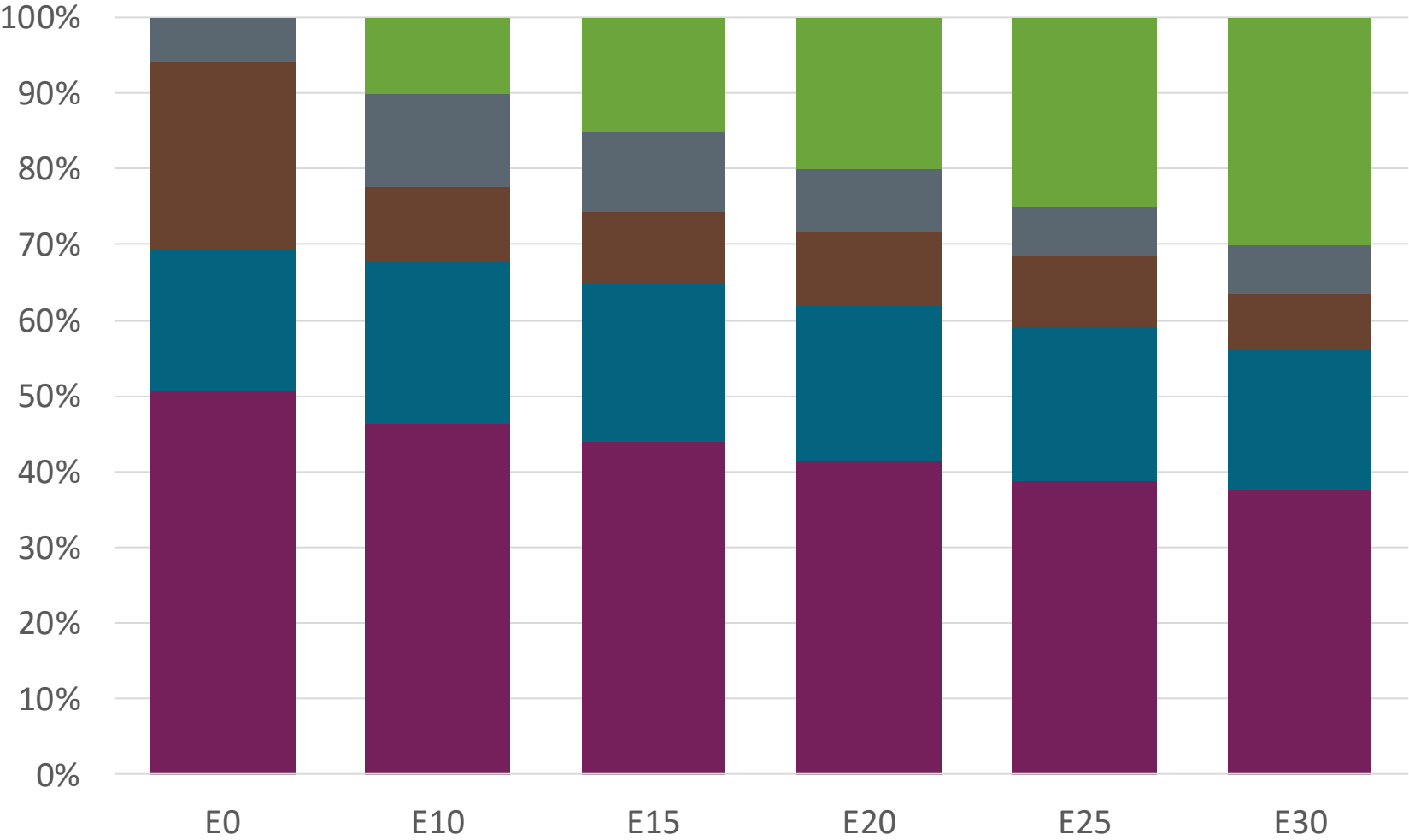
Peru – Regular – Increased Octane



	E0	E10	E15	E20	E25	E30
Octane (RON)	90.0	93.5	95.2	96.9	98.4	105.4
Price (US\$/gal)	\$ 2.410	\$ 2.412	\$ 2.412	\$ 2.412	\$ 2.401	\$ 2.412

Source: Faro90

Peru – Premium – Increased Octane



Octane (RON)	95.0	96.4	97.8	99.4	100.8	102.1
Price (US\$/gal)	\$ 2.623	\$ 2.561	\$ 2.545	\$ 2.531	\$ 2.515	\$ 2.496

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

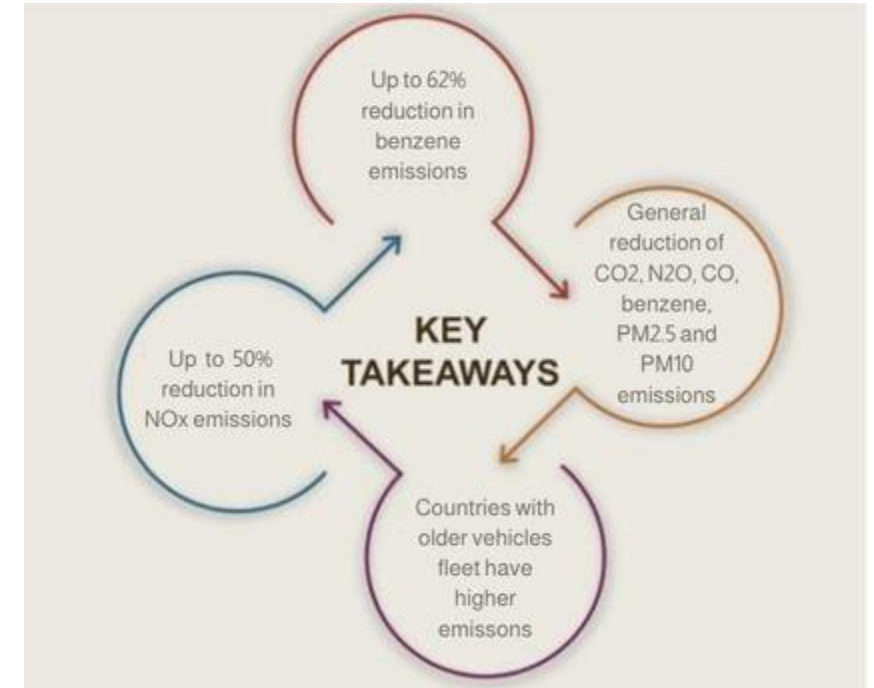
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

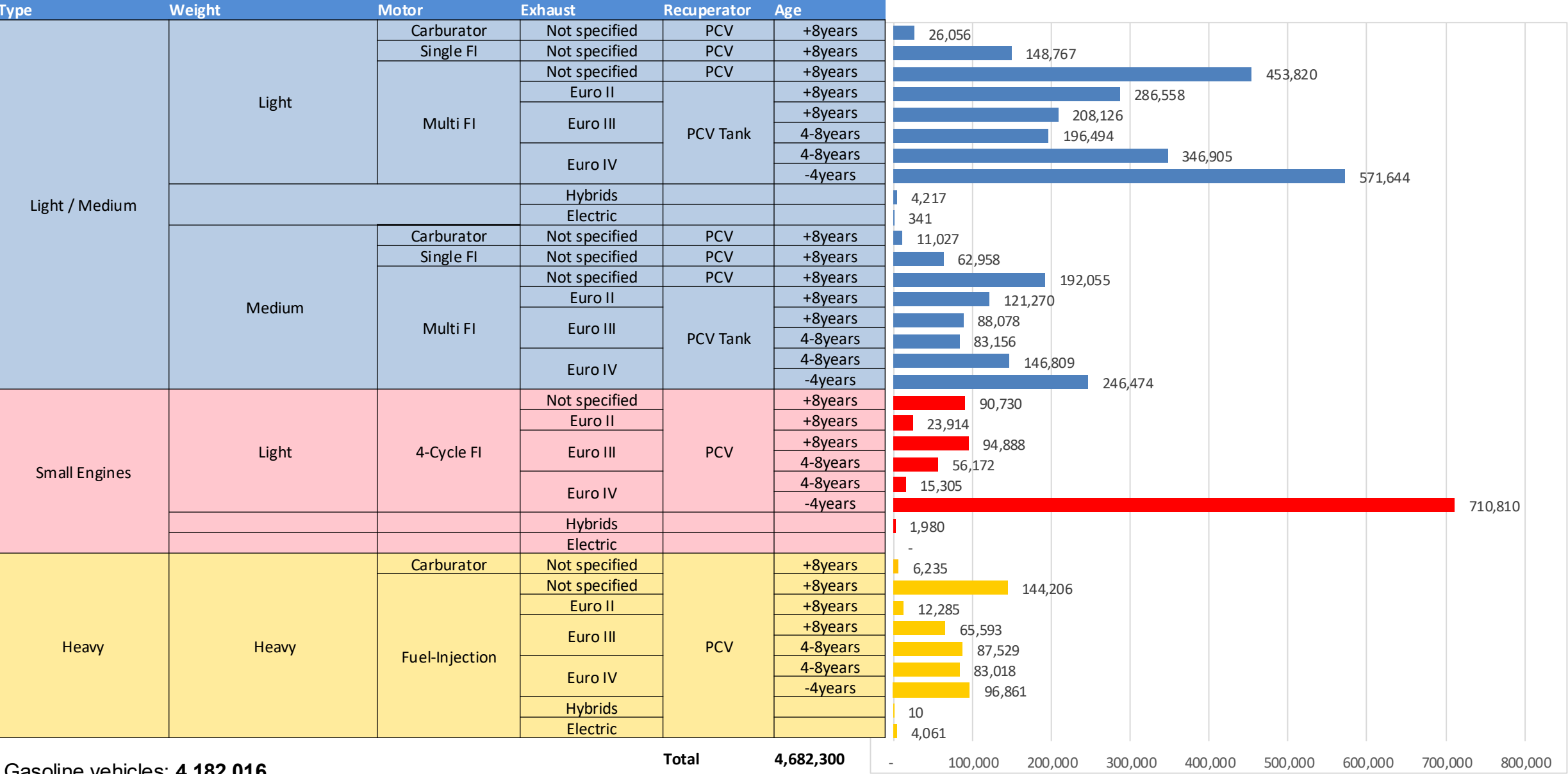
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Peru – Gasoline Vehicle Fleet



Gasoline vehicles: 4,182,016

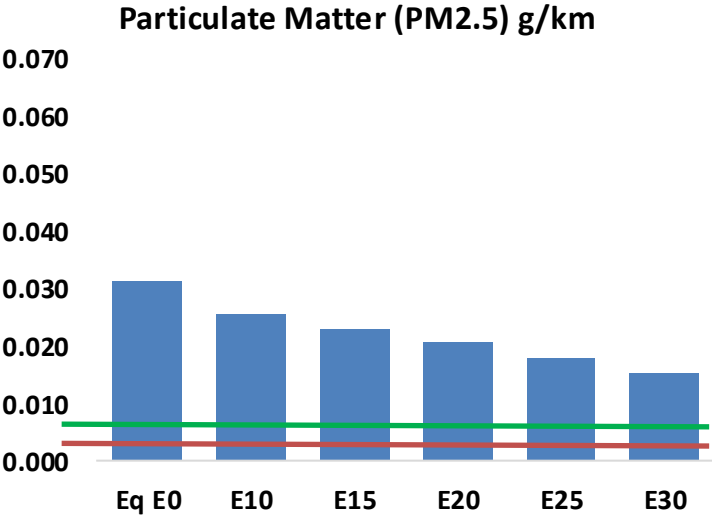
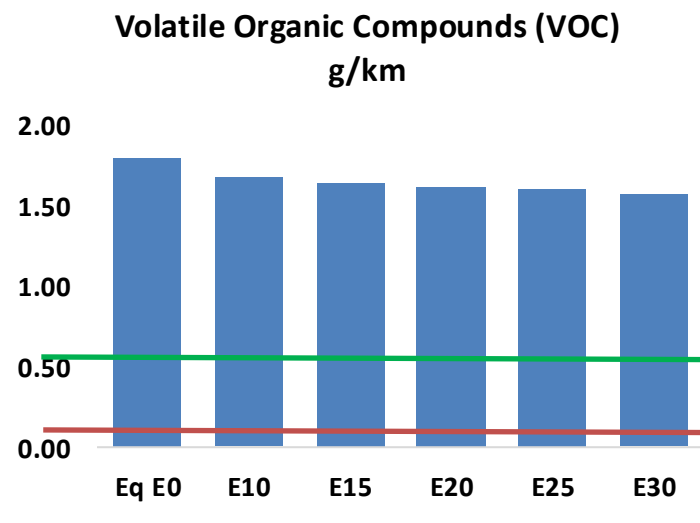
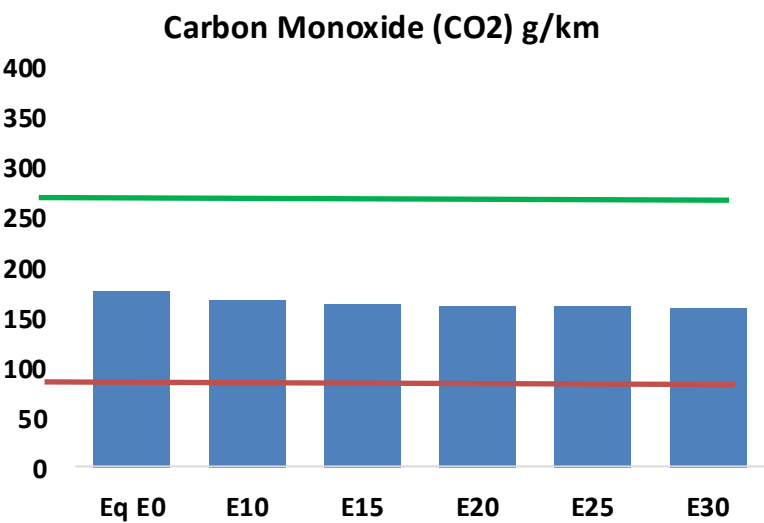
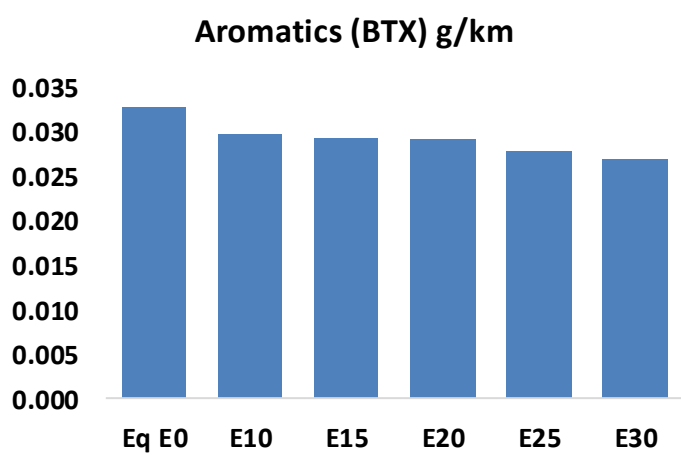
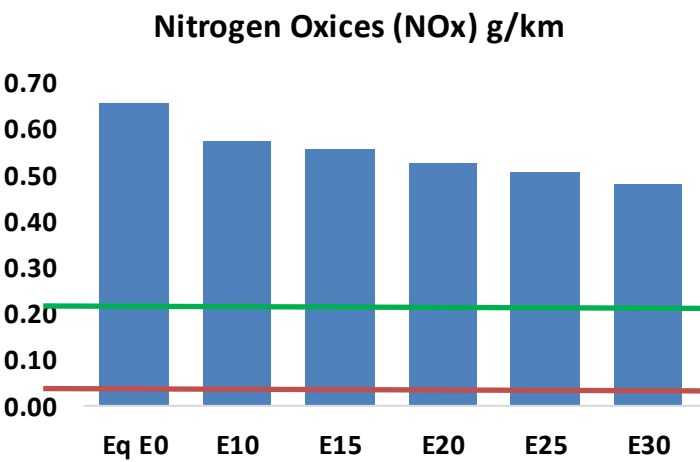
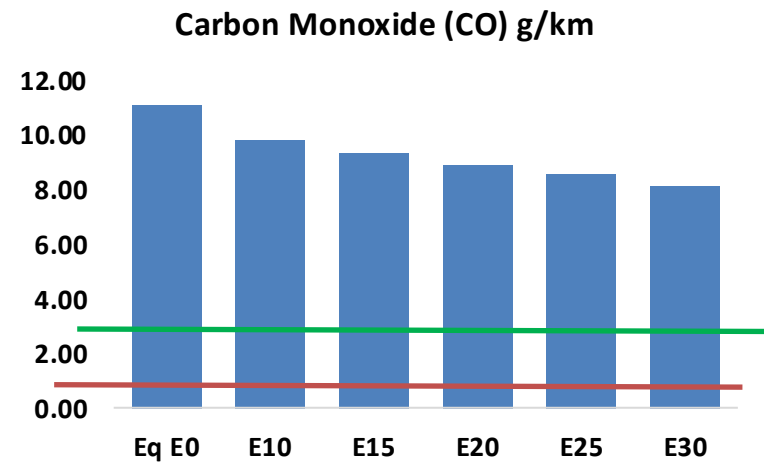
Average Age: 13.2 years

Motorcycles: 21%

Peru – Emissions

United States

Euro 6



Peru – Gasoline Vehicle Fleet Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,170,451	1,104,009	1,037,566	987,711	939,656	903,341	857,311	-11%	-20%
CO2	18,524,884	18,048,530	17,598,640	17,244,815	17,069,728	17,005,418	16,691,959	-5%	-8%
VOC	189,345	183,067	176,788	173,258	170,449	168,514	165,139	-7%	-10%
NOx	69,155	63,853	60,317	58,570	55,695	53,468	50,830	-13%	-19%
SOx	213	197	181	167	154	140	125	-15%	-28%
NH3	6,875	6,807	6,739	6,771	6,847	6,901	6,946	-2%	0%
N2O	190	189	188	189	193	195	198	-1%	2%
CH4	41,845	41,845	41,845	42,472	43,393	44,188	44,941	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	604	579	555	539	525	515	499	-8%	-13%
Acetaldehyde	1,308	1,501	2,202	3,353	4,569	5,221	6,169	68%	249%
Formaldehyde	4,502	4,377	5,102	5,991	6,277	6,944	7,559	13%	39%
Benzene	3,455	3,320	3,146	3,096	3,070	2,936	2,842	-9%	-11%
PM2.5	1,172	1,065	959	865	773	675	572	-18%	-34%
PM10	2,101	1,910	1,719	1,551	1,386	1,210	1,025	-18%	-34%

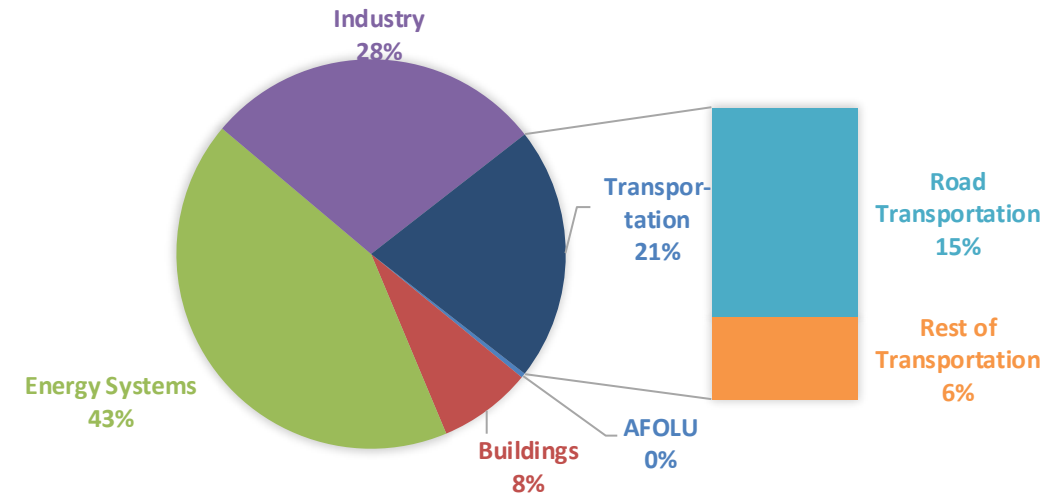
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

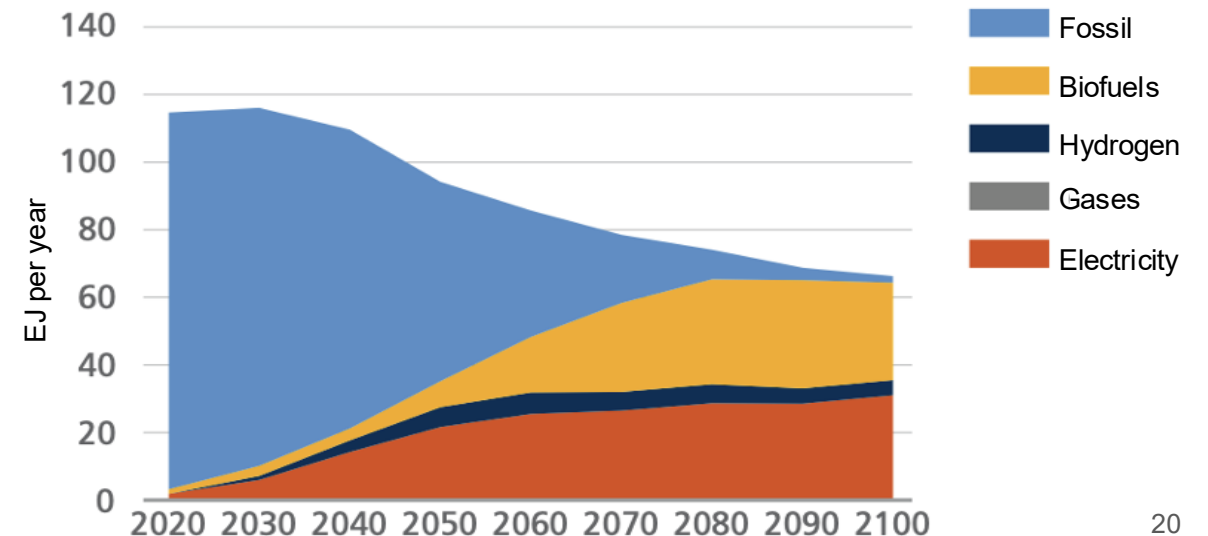
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

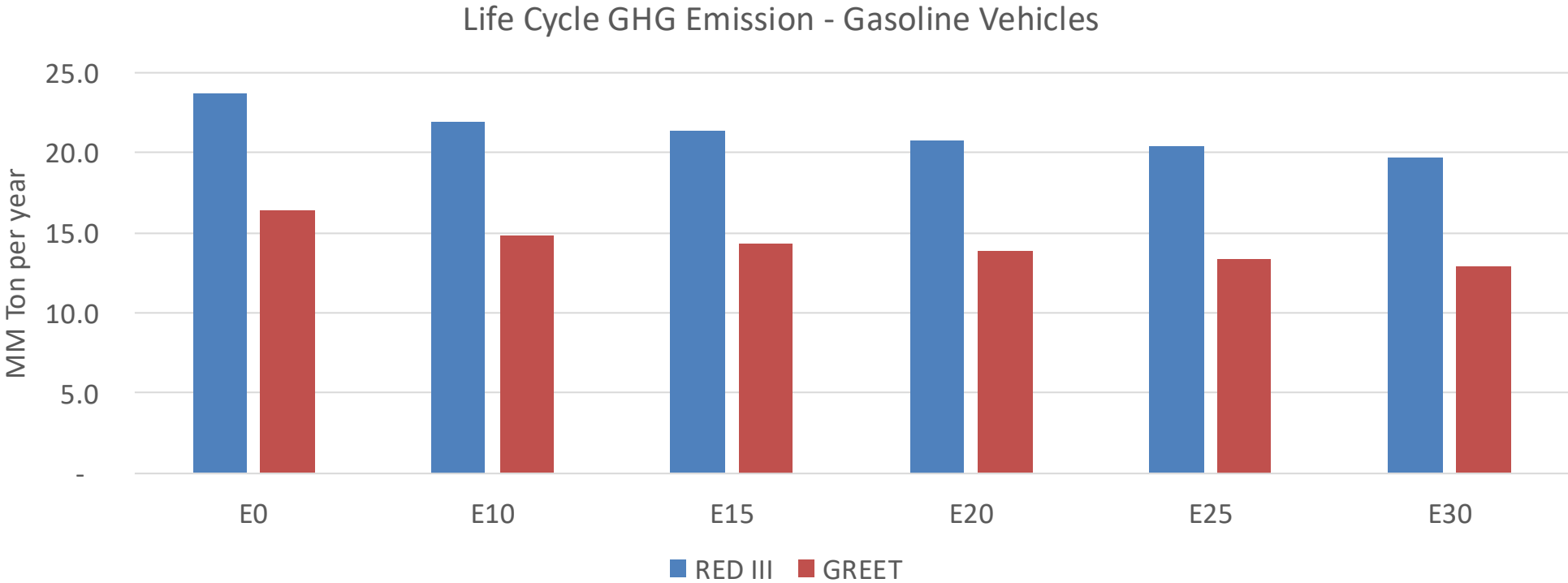
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Peru – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2024 (MMT/year)	111.2
2035 Target (MMT/year)	96.1
Delta	15.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	9.6%	12.5%	15.6%	18.2%	21.6%
RED II/III	0.0%	7.3%	9.5%	12.0%	14.0%	16.5%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	10.4%	13.5%	16.9%	19.7%	23.4%
RED II/III	0.0%	11.4%	14.8%	18.7%	21.8%	25.8%